



T.C. RAMESH
HEAD - TECHNOLOGY,
QUEST GLOBAL

The Internet of Things:

FUNDAMENTAL CHANGE By The Industrial IoT Lies In PROACTIVE OPTIMISATION

There is a shift in revenue generation that is more promising towards the service side than the product side, and the Internet of Things (IoT) would be the driver of this natural extension. T.C. Ramesh, head - technology, Quest Global, speaks with Shanosh Kumar from EFY

Q. What, according to you, is the tipping point that led to the IoT being so popular?

A. We are in the engineering space and have developed highly-engineered devices for many applications. The IoT has been there in different forms from quite some time now. And with all the technology that has come in, it is becoming easier for us to optimise with the Internet. Optimisation is key to the popularity of the IoT.

Q. We have heard of engineering analytics being important to the IoT. Please share your views.

A. Industrial IoT (IIoT) is a smaller segment of the IoT where we talk only about machines. Analytics is an integral part of the IoT, one that cannot be segregated. When it comes to engineering analytics, devices are generating data, which is fed into the swarms of servers that form the Cloud, and analytics is done on that. For example, on an aircraft engine, altitude, speed, thrust, fuel consumption, temperature and pressure data could be downloaded and analysed by the aircraft manufacturer. Analysis is done on two levels: one is predictive analytics based on logistics and the other looks at real data, which is fed back to the system to initiate improvement phases on existing designs.

Q. But analytics has always existed. How has it changed with the IoT?

A. Products have different dimensions. These are no longer standalone and are getting connected with the ecosystem. Though the product life cycle remains the same, what goes into it has changed. It is always better to maintain than repair; this is the benefit we imbibe into the verticles we handle using analytics. We feed data back into machines for improvements. We are training our people in understanding data, using newer tools such as 'R' that are applicable to this business.

Q. Could you share an example of how engineers come up on proactive interventions?

A. While measuring elements like noise, vibration, temperature spin and rotation is when trends emerge. When something starts showing a trend, you can relate it to what might happen. Now, when we analyse trends, we can take interventions to the most possible extent and correct the actions, if required. One is at the sensor level, which is collecting data, and the second is communication. Once data is acquired, you aggregate it, analyse it and then take decisions.

Q. What is the challenge with building IIoT products as opposed to consumer IoT?

A. Consumer electronics can be made intelligent quite effortlessly, but when it comes to industrial products from decades ago that were designed for a specific use case, we need to weigh in the factors and consider whether integrating the IoT is worth the hassle. Assuming that the use case can see phenomenal improvements, we can enable and convert these into intelligent machines. Platforms like ours focus on data and try to create meaningful insight out of it, in order to deliver value in the integral product development segment.

Q. What is important to achieve a breakthrough in the IIoT zone?

A. It is about justifying the return on investments, the disruptions that come across and, more critically, the security of data. Heavy machineries that are involved in power generation, transport, healthcare, automotive, mining and industry equipment are sensitive, not only from the data perspective. Once these major concerns are addressed, we would not have the tendency to broad base the IIoT with commercial devices. Achieving breakthrough is through ensuring reliability in the verticals we play with. **EFY**